

33.0.1 UGC NET CSE | June 2009 | Part 2 | Question: 37

Which of the following changes permission to deny write permission to group and others ?

1. Chmod go-w filex
2. Chmod go w filex
3. Chmod go=w filex
4. None of the above

ugcnetcse-june2009-paper2

33.0.2 UGC NET CSE | December 2004 | Part 2 | Question: 50

The term hacker was originally associated with :

- A. A computer program
- B. Virus
- C. Computer professionals who solved complex computer problems
- D. All of the above

ugcnetcse-dec2004-paper2

Answer key

33.0.3 UGC NET CSE | December 2004 | Part 2 | Question: 46

Data Mining can be used as _____ Tool.

- A. Software B. Hardware C. Research D. Process

ugcnetcse-dec2004-paper2

Answer key

33.0.4 UGC NET CSE | December 2006 | Part 2 | Question: 50

The maximum size of SMS in *IS – 95* is _____ octets.

- A. 120 B. 95 C. 128 D. 64

ugcnetcse-dec2006-paper2

Answer key

33.0.5 UGC NET CSE | December 2006 | Part 2 | Question: 49

Link analysis operation in data mining uses _____ technique.

- A. Classification. B. Association discovery.
C. Visualisation. D. Neural clustering.

ugcnetcse-dec2006-paper2

33.0.6 UGC NET CSE | December 2007 | Part 2 | Question: 50

Identify the incorrect statement :

- A. ATM provides both real time and non-real time service
B. ATM provides faster packet switching than X.25

- C. ATM was developed as part of the work on broadband ISDN
D. ATM does not have application in Non-ISDN environments where very high data rates are required

ugcnetcse-dec2007-paper2

33.0.7 UGC NET CSE | December 2007 | Part 2 | Question: 49



Identify the incorrect statement :

- A. The internet has evolved into phenomenally successful e-commerce engine
B. e-business is synonymous with e-commerce
C. The e-commerce model *B2C* did not begin with billboardware
D. The e-commerce model *G2C* began with billboardware

ugcnetcse-dec2007-paper2

33.0.8 UGC NET CSE | October 2022 | Part 1 | Question: 104



Based on the following passage, answer the Questions :

A 3000 km long trunk operates at 1.536mbps and is used to transmit 64 bytes frames and uses sliding window protocol. The propagation speed is $6\mu\text{sec}/\text{km}$.

The sender window size to get the maximum efficiency is

- A. 108 B. 109 C. 55 D. 56

ugcnetcse-oct2022-paper1

33.0.9 UGC NET CSE | October 2022 | Part 1 | Question: 103



Based on the following passage, answer the Questions :

A 3000 km long trunk operates at 1.536mbps and is used to transmit 64 bytes frames and uses sliding window protocol. The propagation speed is $6\mu\text{sec}/\text{km}$.

The transmission and propagation delays are respectively

- A. $T_t = 333.33\mu\text{sec}$, $T_p = 18000\mu\text{sec}$
B. $T_t = 300\mu\text{sec}$, $T_p = 15360\mu\text{sec}$
C. $T_t = 33.33\mu\text{sec}$, $T_p = 1800\mu\text{sec}$
D. $T_t = 1800\mu\text{sec}$, $T_p = 33.33\mu\text{sec}$

ugcnetcse-oct2022-paper1

Answer key

33.0.10 UGC NET CSE | October 2022 | Part 1 | Question: 102



Based on the following passage, answer the Questions :

A 3000 km long trunk operates at 1.536mbps and is used to transmit 64 bytes frames and uses sliding window protocol. The propagation speed is $6\mu\text{sec}/\text{km}$.

The maximum achievable throughput is

- A. 0.768 B. 0.678 C. 0.901 D. 0.887

ugcnetcse-oct2022-paper1

33.0.11 UGC NET CSE | October 2022 | Part 1 | Question: 101



Based on the following passage, answer the Questions:

A 3000 km long trunk operates at 1.536mbps and is used to transmit 64 bytes frames and uses sliding window

protocol. The propagation speed is $6\mu\text{sec}/\text{km}$.

If only 6 bits are reserved for sequence number field, then the efficiency of the system is

- A. 0.587 B. 0.875 C. 0.578 D. 0.50

ugcnetcse-oct2022-paper1

33.0.12 UGC NET CSE | October 2022 | Part 1 | Question: 100



Based on the following passage, answer the Questions :

A 3000 km long trunk operates at 1.536mbps and is used to transmit 64 bytes frames and uses sliding window protocol. The propagation speed is $6\mu\text{sec}/\text{km}$.

The minimum number of bits required in the sequence number field of the packet is

- A. 6 bits B. 7 bits C. 5 bits D. 4 bits

ugcnetcse-oct2022-paper1

Answer key

33.0.13 UGC NET CSE | October 2022 | Part 1 | Question: 99



Read the following and Answer the questions:

Consider the relational schema of sailors S, Reserves B and Boats B.

Table 1: Sailors S

<i>Sid</i>	<i>Sname</i>	<i>Rating</i>	<i>Age</i>
22	<i>Dustin</i>	7	45.0
29	<i>Brutus</i>	1	33.0
31	<i>Lubber</i>	8	55.5
32	<i>Andy</i>	8	25.5
58	<i>Rusty</i>	10	35.0
64	<i>Horatio</i>	7	35.0
71	<i>Zorba</i>	10	16.0
74	<i>Horatio</i>	9	35.0
85	<i>Art</i>	3	25.5
95	<i>Bob</i>	3	63.5

Table 2: Reserves R

<i>sid</i>	<i>Bid</i>	<i>day</i>
22	101	10/1098
22	108	101098
22	108	101898
22	104	10/798
31	108	11/1098
31	103	11/699
31	104	11/298
64	101	9558
64	102	28858
74	103	98958

Table 3: Boats B

<i>Bid</i>	<i>Bname</i>	<i>Color</i>
101	<i>Interlake</i>	<i>blue</i>
102	<i>Interlake</i>	<i>red</i>
103	<i>Clipper</i>	<i>green</i>
104	<i>Marine</i>	<i>red</i>

Which of the following relational algebra query computes the names of sailor who have reserved all boats?

- A. $\rho(\text{Tempsids}, (\pi \text{ bid Reserves}) / \pi \text{ bid Boats}) \pi \text{ sname} ((\text{Tempsids}) \bowtie \text{Sailors})$
- B. $\rho(\text{Tempsids}, (\pi \text{ sid, bid Reserves}) / \pi \text{ bid Boats}) \pi \text{ sname} ((\text{Tempsids}) \bowtie \text{Sailors})$
- C. $\rho(\text{Tempsids}, (\pi \text{ sid Sailors}) / \pi \text{ bid Boats}) \pi \text{ sname} ((\text{Tempsids}) \bowtie \text{Sailors})$
- D. $\rho(\text{Tempsids}, (\pi \text{ sid Reserves}) / \pi \text{ bid Boats}) \pi \text{ sname} ((\text{Tempsids}) \bowtie \text{Boats})$

ugcnetcse-oct2022-paper1

Answer key 

33.0.14 UGC NET CSE | October 2022 | Part 1 | Question: 98



Read the following and Answer the Questions:

Consider the relational schema of sailors S, Reserves R and Boats B.

Table 1: Sailors S

<i>Sid</i>	<i>Sname</i>	<i>Rating</i>	<i>Age</i>
22	<i>Dustin</i>	7	45.0
29	<i>Brutus</i>	1	33.0
31	<i>Lubber</i>	8	55.5
32	<i>Andy</i>	8	25.5
58	<i>Rusty</i>	10	35.0
64	<i>Horatio</i>	7	35.0
71	<i>Zorba</i>	10	16.0
74	<i>Horatio</i>	9	35.0
85	<i>Art</i>	3	25.5
95	<i>Bob</i>	3	63.5

Table 2: Reserves R

<i>sid</i>	<i>Bid</i>	<i>day</i>
22	101	10/1098
22	108	101098
22	108	101898
22	104	10/798
31	108	11/1098
31	103	11/699
31	104	11/298
64	101	9558
64	102	28858
74	103	98958

Table 3: Boats B

<i>Bid</i>	<i>Bname</i>	<i>Color</i>
101	<i>Interlake</i>	<i>blue</i>
102	<i>Interlake</i>	<i>red</i>
103	<i>Clipper</i>	<i>green</i>
104	<i>Marine</i>	<i>red</i>

Which of the following relational algebra query/queries computes/compute the sid's of sailors with age over 20 who have not reserved red boat?

- A. $\pi_{sid} (\delta_{age > 20} \text{ Sailors }) - \pi_{sid} ((\delta_{color = red} \text{ Boats }) \bowtie \text{Reserves} \bowtie \text{Sailors})$
B. $\pi_{sid} ((\delta_{color \neq red \wedge age > 20} (\text{Boats} \bowtie \text{Sailors} \bowtie \text{Reserves})))$
C. $\pi_{sid} (\delta_{age < 20} \text{ Sailors }) - \pi_{sid} ((\delta_{color = red} \text{ Boats }) \bowtie \text{Reserves} \bowtie \text{Sailors})$
D. $\pi_{sid} (\delta_{age > 20} \text{ Sailors }) \wedge \pi_{sid} ((\delta_{color = red} \text{ Boats }) \bowtie \text{Reserves} \bowtie \text{Sailors})$

ugcnetcse-oct2022-paper1

33.0.15 UGC NET CSE | October 2022 | Part 1 | Question: 97



Read the following and Answer the Questions:

Consider the relational schema of sailors S, Reserves R and Boats B.

Table 1: Sailors S

<i>Sid</i>	<i>Sname</i>	<i>Rating</i>	<i>Age</i>
22	<i>Dustin</i>	7	45.0
29	<i>Brutus</i>	1	33.0
31	<i>Lubber</i>	8	55.5
32	<i>Andy</i>	8	25.5
58	<i>Rusty</i>	10	35.0
64	<i>Horatio</i>	7	35.0
71	<i>Zorba</i>	10	16.0
74	<i>Horatio</i>	9	35.0
85	<i>Art</i>	3	25.5
95	<i>Bob</i>	3	63.5

Table 2: Reserves R

<i>sid</i>	<i>Bid</i>	<i>day</i>
22	101	10/1098
22	108	101098
22	108	101898
22	104	10/798
31	108	11/1098
31	103	11/699
31	104	11/298
64	101	9558
64	102	28858
74	103	98958

Table 3: Boats B

<i>Bid</i>	<i>Bname</i>	<i>Color</i>
101	<i>Interlake</i>	<i>blue</i>
102	<i>Interlake</i>	<i>red</i>
103	<i>Clipper</i>	<i>green</i>
104	<i>Marine</i>	<i>red</i>

Which of the following relational algebra query/queries computes/compute the name of sailors who have reserved red boat?

Q1 π sname ($(\delta_{color = red} Boats) \bowtie Reserves \bowtie Sailors$)

Q2 π sname ($(\pi_{bid} ((\pi_{bid} \delta_{color = red} Boats) \bowtie Reserves) \bowtie Sailors)$)

Q3 π sname ($(\delta_{color = red} Reserves) \bowtie Boats \bowtie Sailors$)

1. Both Q1 and Q3
2. Both Q2 and Q3
3. Only Q1
4. Only Q2

ugcnetcse-oct2022-paper1

33.0.16 UGC NET CSE | October 2022 | Part 1 | Question: 96



Read the following and Answer the Questions:

Consider the relational schema of sailors S, Reserves R and Boats B.

Table 1: Sailors S

<i>Sid</i>	<i>Sname</i>	<i>Rating</i>	<i>Age</i>
22	<i>Dustin</i>	7	45.0
29	<i>Brutus</i>	1	33.0
31	<i>Lubber</i>	8	55.5
32	<i>Andy</i>	8	25.5
58	<i>Rusty</i>	10	35.0
64	<i>Horatio</i>	7	35.0
71	<i>Zorba</i>	10	16.0
74	<i>Horatio</i>	9	35.0
85	<i>Art</i>	3	25.5
95	<i>Bob</i>	3	63.5

Table 2: Reserves R

<i>sid</i>	<i>Bid</i>	<i>day</i>
22	101	10/1098
22	108	101098
22	108	101898
22	104	10/798
31	108	11/1098
31	103	11/699
31	104	11/298
64	101	9558
64	102	28858
74	103	98958

Table 3: Boats B

<i>Bid</i>	<i>Bname</i>	<i>Color</i>
101	<i>Interlake</i>	<i>blue</i>
102	<i>Interlake</i>	<i>red</i>
103	<i>Clipper</i>	<i>green</i>
104	<i>Marine</i>	<i>red</i>

Which of the following relational algebra query/queries computes/compute the name of sailors who have reserved boat 103?

Q1 π sname ($(\sigma_{bid=103} \text{Boats}) \bowtie \text{Sailors}$)

Q2 π sname = $(\sigma_{bid=103} (\text{Reserves} \bowtie \text{Sailors}))$

Q3 π sname = $((\sigma_{bid=103} \text{Reserves}) \bowtie \text{Sailors})$

A. Both Q1 and Q3

B. Both Q2 and Q3

C. Only Q3

D. Only Q2

ugcnetcse-oct2022-paper1

Answer key 

33.0.17 UGC NET CSE | October 2022 | Part 1 | Question: 95



Read the following and Answer the questions:

Consider the relational schema of sailors S, Reserves B and Boats B.

Table 1: Sailors S

<i>Sid</i>	<i>Sname</i>	<i>Rating</i>	<i>Age</i>
22	<i>Dustin</i>	7	45.0
29	<i>Brutus</i>	1	33.0
31	<i>Lubber</i>	8	55.5
32	<i>Andy</i>	8	25.5
58	<i>Rusty</i>	10	35.0
64	<i>Horatio</i>	7	35.0
71	<i>Zorba</i>	10	16.0
74	<i>Horatio</i>	9	35.0
85	<i>Art</i>	3	25.5
95	<i>Bob</i>	3	63.5

Table 2: Reserves R

<i>sid</i>	<i>Bid</i>	<i>day</i>
22	101	10/1098
22	108	101098
22	108	101898
22	104	10/798
31	108	11/1098
31	103	11/699
31	104	11/298
64	101	9558
64	102	28858
74	103	98958

Table 3: Boats B

<i>Bid</i>	<i>Bname</i>	<i>Color</i>
101	<i>Interlake</i>	<i>blue</i>
102	<i>Interlake</i>	<i>red</i>
103	<i>Clipper</i>	<i>green</i>
104	<i>Marine</i>	<i>red</i>

Which of the following relational algebra query computes the names of sailors who have reserved a red and a green boat?

- A. ρ (Tempred, π sid $((\delta_{\text{color}} = \text{'red'}$ Boats) $\bowtie$ Reserves)), ρ (Tempgreen, π sid $((\delta_{\text{color}} = \text{'green'}$ Boats) $\bowtie$ Reserves)), π sname (Tempred $\cap$ Tempgreen) $\bowtie$ Sailors
- B. ρ (Tempboats 2, $(\delta_{\text{color}} = \text{'red'}$ Boats) $\cap$ $(\delta_{\text{color}} = \text{'green'}$ Boats)) π sname (Tempboats 2 $\bowtie$ Reserves)
- C. π sname $((\delta_{\text{color}} = \text{'red'}$ Boats) $\cap$ $(\delta_{\text{color}} = \text{'green'}$ Boats)) (Tempboats $\bowtie$ Sailors $\bowtie$ Reserves)
- D. ρ (Tempboats 2, $(\delta_{\text{color}} = \text{red}$ Boats $\delta_{\text{color}} = \text{'green'}$ Boats)) π sname (Tempboats 2 $\bowtie$ sailors)

ugcnetcse-oct2022-paper1

33.0.18 UGC NET CSE | October 2022 | Part 1 | Question: 94



In a game playing search tree, upto which depth $\alpha - \beta$ pruning can be applied?

- (A) Root (0) level
- (B) 6 level
- (C) 8 level
- (D) Depends on utility value in a breadth first order

Choose the correct answer from the options given below :

- A. (B) and (C) only B. (A) and (B) only C. (A) (B) and (C) only D. (A) and (D) only

ugcnetcse-oct2022-paper1

Answer key

33.0.19 UGC NET CSE | October 2022 | Part 1 | Question: 93



Consider the traversal of a tree

Preorder $\rightarrow$ ABCEIFJDGHKL

Inorder $\rightarrow$ EICFJBGDKHLA

Which of the following is correct post order traversal?

- A. EIFJCKGLHDBA
- C. FCGKLHDBAELJ

- B. FCGKLHDBUAE
- D. IEJFCGKLHDBA

ugcnetcse-oct2022-paper1

33.0.20 UGC NET CSE | October 2022 | Part 1 | Question: 92



A good software requirement specification does NOT have the characteristic

- A. Completeness
- B. Consistency
- C. Clarity
- D. Reliability

ugcnetcse-oct2022-paper1

Answer key

33.0.21 UGC NET CSE | October 2022 | Part 1 | Question: 91



Let $R(ABCDEFGH)$ be a relation schema and F be the set of dependencies $F = \{A \rightarrow B, ABCD \rightarrow E, EF \rightarrow G, EF \rightarrow H \text{ and } ACDF \rightarrow EG\}$. The minimal cover of a set of functional dependencies is

- A. $A \rightarrow B, ACD \rightarrow E, EF \rightarrow G$, and $EF \rightarrow H$
- B. $A \rightarrow B, ACD \rightarrow E, EF \rightarrow G, EF \rightarrow H$ and $ACDF \rightarrow G$
- C. $A \rightarrow B, ACD \rightarrow E, EF \rightarrow G, EF \rightarrow H$ and $ACDF \rightarrow E$
- D. $A \rightarrow B, ABCD \rightarrow E, EF \rightarrow H$ and $EF \rightarrow G$

ugcnetcse-oct2022-paper1

33.0.22 UGC NET CSE | October 2022 | Part 1 | Question: 90



Modules X and Y operate on the same input and output, then the cohesion is

- A. Logical cohesion
- B. Sequential cohesion
- C. Procedural cohesion
- D. Communicational cohesion

ugcnetcse-oct2022-paper1

33.0.23 UGC NET CSE | October 2022 | Part 1 | Question: 89



For the following page reference string 4, 3, 2, 1, 4, 3, 5, 4, 3, 2, 1, 5, the number of page faults that occur in Least Recently Used (LRU) page replacement algorithm with frame size 3 is

- A. 6
- B. 8
- C. 10
- D. 12

ugcnetcse-oct2022-paper1

33.0.24 UGC NET CSE | October 2022 | Part 1 | Question: 88



Given below are two statements :

Statement I: "Grandparent is a parent of one's parent".

Statement II: First Order Predicate Logic (FOPL) representation of above statement is

$\forall g, c \text{ grandparent } (g, c) \Leftrightarrow \exists p \text{ parent } (g, p) \wedge \text{parent } (p, c)$

In the light of the above statements, choose the most appropriate answer from the options given below :

- A. Both Statement I and Statement II are correct
- B. Both Statement I and Statement II are incorrect
- C. Statement I is correct but Statement II is incorrect
- D. Statement I is incorrect but Statement II is correct

ugcnetcse-oct2022-paper1



Match **List I** with **List II** :

List I	List II
(A) Localization	(I) Encapsulation
(B) Packaging or binding of a collection of items	(II) Abstraction
(C) Mechanism that enables designer to focus on essential details of a program component.	(III) Characteristic of software that indicates the manner in which information is concentrated in program
(D) Information hiding	(IV) Suppressing the operational details of a program component

Choose the correct answer from the options given below :

- A. (A)-(I), (B)-(II), (C)-(III), (D)-(IV) B. (A)-(II), (B)-(I), (C)-(III), (D)-(IV)
 C. (A)-(III), (B)-(I), (C)-(II), (D)-(IV) D. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

ugcnetcse-oct2022-paper1



The total storage capacity of a floppy disk having 80 tracks and storing 128 bytes/sector is 163,840 bytes. How many sectors does this disk have?

- A. 2^7 B. 2048 C. 4 K D. 16

ugcnetcse-oct2022-paper1

Answer key



The process to gather the software requirements from client, analyze and document is known as -

- A. Software Engineering Process B. User Engineering Process
 C. Requirement Elicitation Process D. Requirement Engineering Process

ugcnetcse-oct2022-paper1



If every requirement can be checked by a cost-effective process, then SRS is called

- A. Verifiable B. Traceable C. Modifiable D. Complete

ugcnetcse-oct2022-paper1



RAD software process model stands for

- A. Rapid Application Development B. Relative Application Development
 C. Rapid Application Design D. Recent Application Development,

ugcnetcse-oct2022-paper1

Answer key

33.0.30 UGC NET CSE | October 2022 | Part 1 | Question: 79

The reduced grammar equivalent to the grammar, whose production rules are given below, is

$$S \rightarrow AB \mid CA$$

$$B \rightarrow BC \mid AB$$

$$A \rightarrow a$$

$$C \rightarrow aB \mid b$$

- A. $S \rightarrow CA, A \rightarrow a, C \rightarrow b$
 B. $S \rightarrow CA \mid B, B \rightarrow BC \mid B, A \rightarrow a, C \rightarrow aB \mid b$
 C. $S \rightarrow CA \mid B, B \rightarrow BC, A \rightarrow a, C \rightarrow aB \mid b$
 D. $S \rightarrow AB \mid AC, B \rightarrow BC \mid BA, A \rightarrow a, C \rightarrow aB \mid b$

ugcnetcse-oct2022-paper1

33.0.31 UGC NET CSE | October 2022 | Part 1 | Question: 78

A magnetic tape drive has transport speed of 200 inches per second and a recording density of 1600 bytes per inch. The time required to write 600000 bytes of data grouped in 100 characters record with a blocking factor 10 is

- A. 2.0625sec
 B. 2.6251sec
 C. 2.0062sec
 D. 2.6150sec

ugcnetcse-oct2022-paper1

33.0.32 UGC NET CSE | October 2022 | Part 1 | Question: 77

Consider $L = a^*b, a^*a, b^*a^*$

Which of the following string is NOT in L^* ?

- A. baaaaabaaaaa
 B. abaabaaabaa
 C. aaaabaaaa
 D. baaaabaa

ugcnetcse-oct2022-paper1

Answer key

33.0.33 UGC NET CSE | October 2022 | Part 1 | Question: 76

Consider the following statements:

Statement I: LALR parser is more powerful than canonical LR Parser.

Statement II: SLR parser is more powerful than LALR

Which of the following is correct?

- A. Statement I true and Statement II false
 B. Statement I false and Statement II true
 C. Both Statement I and Statement II false
 D. Both Statement I and Statement II true

ugcnetcse-oct2022-paper1

Answer key

33.0.34 UGC NET CSE | October 2022 | Part 1 | Question: 75

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R) :

Assertion (A): $\bar{p}$

Reason (R): $(r \rightarrow q, r \vee s, s \rightarrow q, p \rightarrow q)$

In the light of the above statements, choose the correct answer from the options given below :

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
- B. Both (A) and (R) are true but (R) is (NOT) the correct explanation of (A)
- C. (A) is true but (R) is false
- D. (A) false but (R) is true

ugcnetcse-oct2022-paper1

33.0.35 UGC NET CSE | October 2022 | Part 1 | Question: 73



Match List I with List II:

List I

List II

- | | |
|------------------------|--|
| (A) Physical layer | (I) Routing of the signals divide the outgoing message into packets, to act as network controller for routing data |
| (B) Data link layer | (II) Make and break connections, define voltages and data rates convert data bits into electrical signal |
| (C) Network layer | (III) Synchronization, error detection and correction. To assemble outgoing message into frames |
| (D) Presentation layer | (IV) It works as a translating layer |

Choose the correct answer from the options given below :

- | | |
|---|---|
| A. (A)-(IV), (B)-(III), (C)-(II), (D)-(I) | B. (A)-(II), (B)-(III), (C)-(IV), (D)-(I) |
| C. (A)-(IV), (B)-(III), (C)-(I), (D)-(II) | D. (A)-(II), (B)-(III), (C)-(I), (D)-(IV) |

ugcnetcse-oct2022-paper1

33.0.36 UGC NET CSE | October 2022 | Part 1 | Question: 72



Consider the following:

List I

List II

- | | |
|-----------------------------------|------------------------|
| (A) Stack algorithm | (I) Deadlock |
| (B) Elevator algorithm | (II) Disk scheduling |
| (C) Priority scheduling algorithm | (III) Page replacement |
| (D) Havender's algorithm | (IV) CPU scheduling |

Which of the following is correct matching?

- | | |
|---|---|
| A. (A)-(III), (B)-(II), (C)-(IV), (D)-(I) | B. (A)-(II), (B)-(III), (C)-(IV), (D)-(I) |
| C. (A)-(III), (B)-(II), (C)-(I), (D)-(IV) | D. (A)-(II), (B)-(III), (C)-(I), (D)-(IV) |

ugcnetcse-oct2022-paper1

33.0.37 UGC NET CSE | October 2022 | Part 1 | Question: 71



Which statement is false?

- A. All function calls in C pass arguments using call by value.
- B. Call by reference enables a called function to modify a variable in calling function.
- C. Call by value is always more efficient than call by reference.
- D. Programmers use pointers and indirection operation to simulate call by reference.

ugcnetcse-oct2022-paper1

33.0.38 UGC NET CSE | October 2022 | Part 1 | Question: 70

Of the following, which is NOT a logical error?

- A. Using the ' $' == '$ ' instead of ' $' = '$ ' to determine if two values are equal,
- B. Divide by zero
- C. Failing to initialize counter and total variables before the body of loop,
- D. Using commas instead of two required semicolon in a for loop header

ugcnetcse-oct2022-paper1

Answer key

33.0.39 UGC NET CSE | October 2022 | Part 1 | Question: 69

Let $e = 0.0005$, and Let R_e be the relation $\{(x, y) \in R^2 : |x - y| < e\}$. R_e could be interpreted as the relation approximately equal. R_e is

- (A) Reflexive
- (B) Symmetric
- (C) transitive

Choose the correct answer from the options given below :

- A. (A) and (B) only true
- B. (B) and (C) only true
- C. (A) and (C) only true
- D. (A), (B) and (C) true

ugcnetcse-oct2022-paper1

33.0.40 UGC NET CSE | October 2022 | Part 1 | Question: 68

Consider the following two lists :

List I

List II

- | | |
|--------------------|------------------------------|
| (A) Stack overflow | (I) Software interrupt |
| (B) Timer | (II) Internal interrupt |
| (C) Invalid opcode | (III) External interrupt |
| (D) Superior call | (IV) Machine check interrupt |

Which of the following is correct match ?

- A. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
- B. (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
- C. (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
- D. (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

ugcnetcse-oct2022-paper1

33.0.41 UGC NET CSE | October 2022 | Part 1 | Question: 67

Consider the following related to Fourth Generation Technique (4GT) :

- (A) It controls efforts.
- (B) It controls resources.
- (C) It controls cost of development.

Choose the correct answer from the options given below :

- A. (A) and (B) only
- B. (B) and (C) only
- C. (C) and (A) only
- D. All (A), (B) and (C)

ugcnetcse-oct2022-paper1

33.0.42 UGC NET CSE | October 2022 | Part 1 | Question: 66

Which mechanism in XML allows organizations to specify globally unique names as element tags in documents?

- A. root B. header C. schema D. namespace

ugcnetcse-oct2022-paper1

33.0.43 UGC NET CSE | October 2022 | Part 1 | Question: 65

This transformation is called

$$\begin{bmatrix} \bar{x} \\ \bar{y} \\ \bar{z} \\ \bar{w} \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ e & f & g & h \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- A. Scaling B. Shear C. Homography D. Steganography

ugcnetcse-oct2022-paper1

33.0.44 UGC NET CSE | October 2022 | Part 1 | Question: 64

Match List I with List II :

List I List II

- (A) BIND (I) Block the caller until a connection attempt arrives
(B) LISTEN (II) Give a local address to a socket
(C) ACCEPT (III) Show willingness to accept connections
(D) SOCKET (IV) Create a new point

Choose the correct answer from the options given below :

- A. (A)-(I), (B)-(III), (C)-(II), (D)-(IV) B. (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
C. (A)-(III), (B)-(II), (C)-(I), (D)-(IV) D. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

ugcnetcse-oct2022-paper1

33.0.45 UGC NET CSE | October 2022 | Part 1 | Question: 63

How many rotations are required during the construction of an AVL tree if the following elements are to be added in the given sequence?

35, 50, 40, 25, 30, 60, 78, 20, 28

- A. 2 left rotations, 2 right rotations B. 2 left rotations, 3 right rotations
C. 3 left rotations, 2 right rotations D. 3 left rotations, 1 right rotation

ugcnetcse-oct2022-paper1

Answer key

33.0.46 UGC NET CSE | October 2022 | Part 1 | Question: 62

Match List I with List II :

List I

List II

- | | |
|------------------------|--|
| (A) Firmware | (I) Number of logical records into physical blocks |
| (B) Batch file | (II) ASCII format |
| (C) Packing | (III) Resource allocation |
| (D) Banker's Algorithm | (IV) ROM |

Choose the correct answer from the options given below :

- | | |
|---|---|
| A. (A)-(II), (B)-(I), (C)-(IV), (D)-(III) | B. (A)-(II), (B)-(I), (C)-(III), (D)-(IV) |
| C. (A)-(IV), (B)-(II), (C)-(I), (D)-(III) | D. (A)-(IV), (B)-(I), (C)-(II), (D)-(III) |

ugcnetcse-oct2022-paper1

33.0.47 UGC NET CSE | October 2022 | Part 1 | Question: 61



Consider the production rules of grammar G :

$$\begin{aligned} S &\rightarrow AbB \\ A &\rightarrow aAb \mid \lambda \\ B &\rightarrow bB \mid \lambda \end{aligned}$$

Which of the following language L is generated by grammar G ?

- | | |
|--|---|
| A. $L = \{a^n b^m : n \geq 0, m > n\}$ | B. $L = \{a^n b^m : n \geq 0, m \geq 0\}$ |
| C. $L = \{a^n b^m : n \geq m\}$ | D. $L = \{a^n b^m : n \geq m, m > 0\}$ |

ugcnetcse-oct2022-paper1

33.0.48 UGC NET CSE | October 2022 | Part 1 | Question: 58



A trigger is

- A. A statement that enables to start DBMS
- B. A statement that is executed by the user when debugging an application program.
- C. A condition the system tests for the validity of the database user.
- D. A statement that is executed automatically by the system as a side effect of modification to the database.

ugcnetcse-oct2022-paper1

33.0.49 UGC NET CSE | October 2022 | Part 1 | Question: 57



Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): A load-and-go assembler avoids the overhead of writing the object program out and reading it back in.
Reason (R): This can be done with either one-pass or two pass assembler.

In the light of the above statements, choose the correct answer from the options given below :

- A. Both (A) and (R) are true and (R) is the correct explanation of (A),
- B. Both (A) and (R) are true but (R) is (NOT) the correct explanation of (A)
- C. (A) is true but (R) is false
- D. (A) is false but (R) is true

ugcnetcse-oct2022-paper1

ugcnetcse-oct2022-paper1 hamming-code



ugcnetcse-oct2022-paper1

ugcnetcse-oct2022-paper1

ugcnetcse-oct2022-paper1



A. X is true

C. Both X and Y are true

D. It does not depend on X and Y

ugcnetcse-oct2022-paper1

33.0.55 UGC NET CSE | October 2022 | Part 1 | Question: 49



If an operating system does not allow a child process to exist when the parent process has been terminated, this phenomenon is called as.

A. Threading

B. Cascading termination

C. Zombie termination

D. Process killing

ugcnetcse-oct2022-paper1

33.0.56 UGC NET CSE | October 2022 | Part 1 | Question: 48



Pointers cannot be used to

A. find the address of a variable in memory

B. reference value directly

C. simulate call by reference

D. manipulate dynamic data structure

ugcnetcse-oct2022-paper1

33.0.57 UGC NET CSE | October 2022 | Part 1 | Question: 47



In a database, a rule is defined as $(P1 \text{ and } P2) \text{ or } P3 \rightarrow R1(0.8) \text{ and } R2(0.3)$, where $P1, P2, P3$ are premises and $R1, R2$ are conclusions of rules with certainty factors (CF) 0.8 and 0.3 respectively. If any running program has produced $P1, P2, P3$ with CF as 0.5, 0.8, 0.2 respectively. find the CF of results on the basis of premises.

A. $CF(R1 = 0.8), CF(R2 = 0.3)$

B. $CF(R1 = 0.40), CF(R2 = 0.15)$

C. $CF(R1 = 0.15), CF(R2 = 0.35)$

D. $CF(R1 = 0.8), CF(R2 = 0.35)$

ugcnetcse-oct2022-paper1

33.0.58 UGC NET CSE | October 2022 | Part 1 | Question: 46



A top down approach to programming calls for :

Statement I: Working from the general to the specific.

Statement II: Postpone the minor decisions.

Statement III: A systematic approach.

Statement IV: Intermediate coding of the problem

Which of the following is true?

A. Statement I only

B. Statement I and Statement II only

C. Statement I, Statement II and Statement III only

D. Statement I, Statement II and Statement IV only

ugcnetcse-oct2022-paper1

33.0.59 UGC NET CSE | October 2022 | Part 1 | Question: 45



Match **List I** with **List II** :

List I	List II
(A) Type 0	(I) Finite automata
(B) Type 1	(II) Turing machine
(C) Type 2	(III) Linear bound automata
(D) Type 3	(IV) Pushdown automata

Choose the correct answer from the options given below :

- A. (A) – (III), (B) – (IV), (C) – (II), (D) – (I)
 C. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

- B. (A) – (II), (B) – (III), (C) – (IV), (D) – (I)
 D. (A) – (II), (B) – (III), (C) – (II), (D) – (IV)

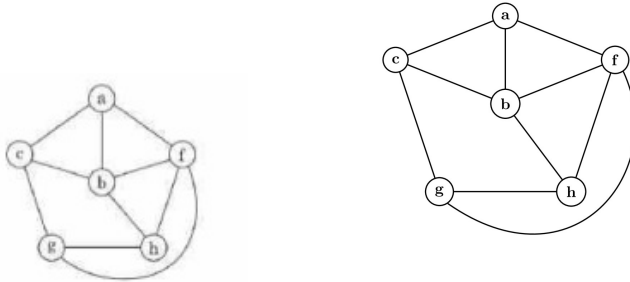
ugcnetcse-oct2022-paper1

Answer key

33.0.60 UGC NET CSE | October 2022 | Part 1 | Question: 44



Consider the following graph :



For the graph; the following sequences of depth first search (DFS) are given

- (A) abcghf
 (B) abfchg
 (C) abfhgc
 (D) afghbc

Which of the following is correct?

- A. (A), (B) and (D) only
 C. (B), (C) and (D) only
 B. (A), (B), (C) and (D)
 D. (A), (C) and (D) only

ugcnetcse-oct2022-paper1

33.0.61 UGC NET CSE | October 2022 | Part 1 | Question: 43



Hidden surface removal problem with minimal 3D pipeline can be solved with

- A. Painter's algorithm
 C. Brute force rasterization algorithm
 B. Window Clipping algorithm
 D. Flood fill algorithm

ugcnetcse-oct2022-paper1

Answer key

33.0.62 UGC NET CSE | October 2022 | Part 1 | Question: 42



Which one is a connectionless transport - layer protocol that belongs to the Internet protocol family?

- A. Transmission Control Protocol (TCP)
 C. Routing Protocol (RP)
 B. User Datagram Protocol (UDP)
 D. Datagram Control Protocol (DCP),

ugcnetcse-oct2022-paper1

33.0.63 UGC NET CSE | October 2022 | Part 1 | Question: 41



Let $(\{a, b\}, *)$ be a semigroup, where $a * a = b$.

- (A) $a * b = b * a$
 (B) $b * b = b$

Choose the most appropriate answer from the options given below :

- A. (A) only true
C. Both (A) and (B) true,

- B. (B) only true
D. Neither (A) nor (B) true,

ugcnetcse-oct2022-paper1

Answer key

33.0.64 UGC NET CSE | October 2022 | Part 1 | Question: 40



The number of gate inputs, required to realize expression $AB\bar{C} + A\bar{B}CD + E\bar{F} + AD$ is

- A. 12 B. 13 C. 14 D. 15

ugcnetcse-oct2022-paper1

33.0.65 UGC NET CSE | October 2022 | Part 1 | Question: 39



There are three boxes. First box has 2 white, 3 black and 4 red balls. Second box has 3 white, 2 black and 2 red balls. Third box has 4 white, 1 black and 3 red balls. A box is chosen at random and 2 balls are drawn out of which 1 is white, and 1 is red. What is the probability that the balls came from first box?

- A. 0.237 B. 0.723 C. 0.18 D. 0.452

ugcnetcse-oct2022-paper1

33.0.66 UGC NET CSE | October 2022 | Part 1 | Question: 38



Match **List I** with **List II**:

List I	List II
(A) DES	(I) Key size - 256
(B) AES	(II) Key size - 1024
(C) 3 DES	(III) Key size - 56
(D) RSA	(IV) Key size - 168

- A. (A) – (I), (B) – (II), (C) – (IV), (D) – (III)
B. (A) – (III), (B) – (I), (C) – (IV), (D) – (II)
C. (A) – (III), (B) – (IV), (C) – (II), (D) – (I)
D. (A) – (IV), (B) – (II), (C) – (III), (D) – (I)

ugcnetcse-oct2022-paper1

Answer key

33.0.67 UGC NET CSE | October 2022 | Part 1 | Question: 37



The logic expression $(\bar{P} \wedge Q) \vee (P \wedge Q) \vee (P \wedge \bar{Q})$ is equivalent to

- A. $\bar{P} \vee Q$, B. $P \vee \bar{Q}$ C. $P \vee Q$ D. $\bar{P} \vee \bar{Q}$

ugcnetcse-oct2022-paper1

Answer key

33.0.68 UGC NET CSE | October 2022 | Part 1 | Question: 36



Consider an error-free 64kbps satellite channel used to send 512 byte data frames in one direction with very short acknowledgements coming back the other way. What is the maximum throughput for window size of 15?

- A. 32kbps B. 48kbps
C. 64kbps D. 70kbps,

ugcnetcse-oct2022-paper1

33.0.69 UGC NET CSE | October 2022 | Part 1 | Question: 35



The condition $\text{num}! = 65$ cannot be replaced by

- A. $\text{num} > 65 \parallel \text{num} < 65$ B. $!(\text{num} == 65)$
C. $\text{num} - 65$ D. $!(\text{num} - 65)$

Answer key **33.0.70 UGC NET CSE | October 2022 | Part 1 | Question: 34**

The model in which the requirements are implemented by its category is

- A. Evolutionary Development Model
 B. Waterfall Model
 C. Prototyping Model
 D. Iterative Enhancement Model

ugcnetcse-oct2022-paper1

33.0.71 UGC NET CSE | October 2022 | Part 1 | Question: 33

Which of the following is correct for the destination address 4 A : 30 : 10 : 21 : 10 : 1 A ?

- A. unicast address
 B. multicast address
 C. broadcast address
 D. unicast and broadcast address

ugcnetcse-oct2022-paper1

Answer key **33.0.72 UGC NET CSE | October 2022 | Part 1 | Question: 32**

Consider the properties of recursively enumerable sets :

- (A) Finiteness
 (B) Context Freedom
 (C) Emptiness

Which of the following is true?

- A. Only (A) and (B) are not decidable
 B. Only (B) and (C) are not decidable
 C. Only (C) and (A) are not decidable,
 D. All (A), (B) and (C) are not decidable

ugcnetcse-oct2022-paper1

33.0.73 UGC NET CSE | October 2022 | Part 1 | Question: 31

Alpha and Beta testing are forms of

- A. White-Box Testing,
 B. Black-Box Testing
 C. Acceptance Testing
 D. System Testing

ugcnetcse-oct2022-paper1

33.0.74 UGC NET CSE | October 2022 | Part 1 | Question: 30

Consider the following:

Which of the following is correct matching ?

List I	List II
(A) Activation record	(I) Linking Loader
(B) Location counter	(II) Garbage
(C) Reference count	(III) Subroutine Call
(D) Address relocation	(IV) Assembler

- A. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)
 B. (A) – (IV), (B) – (III), (C) – (I), (D) – (II)
 C. (A) – (IV), (B) – (III), (C) – (II), (D) – (I)
 D. (A) – (II), (B) – (III), (C) – (I), (D) – (IV)

ugcnetcse-oct2022-paper1

33.0.75 UGC NET CSE | October 2022 | Part 1 | Question: 29

Which of the following is correct for the destination address 4 A : 30 : 10 : 21 : 10 : 1 A ?

- A. unicast address
- B. multicast address
- C. broadcast address
- D. unicast and broadcast address

ugcnetcse-oct2022-paper1

Answer key

33.0.76 UGC NET CSE | October 2022 | Part 1 | Question: 28

Consider the properties of recursively enumerable sets :

- (A) Finiteness
- (B) Context Freedom
- (C) Emptiness

Which of the following is true?

- A. Only (A) and (B) are not decidable
- B. Only (B) and (C) are not decidable
- C. Only (C) and (A) are not decidable,
- D. All (A), (B) and (C) are not decidable

ugcnetcse-oct2022-paper1

Answer key

33.0.77 UGC NET CSE | October 2022 | Part 1 | Question: 27

Alpha and Beta testing are forms of

- A. White-Box Testing
- B. Black-Box Testing
- C. Acceptance Testing
- D. System Testing

ugcnetcse-oct2022-paper1

Answer key

33.0.78 UGC NET CSE | October 2022 | Part 1 | Question: 26

Consider the following :

List I

- (A) Activation record
- (B) Location counter
- (C) Reference count
- (D) Address relocation

List II

- (I) Linking Loader
- (II) Garbage Collection
- (III) Subroutine Call
- (IV) Assembler

Which of the following is correct matching ?

- A. (A)-(III), (B)-(IV), (C) -(I), (D)-(II)
- B. (A) -(IV), (B) - (III), (C)- (I), (D) - (II)
- C. (A)-(IV), (B) - (III), (C) - (II), (D) - (I)
- D. (A)-(II), (B) - (III), (C) - (I), (D) - (IV)

ugcnetcse-oct2022-paper1

Answer key

33.0.79 UGC NET CSE | October 2022 | Part 1 | Question: 25

Which mode is a block cipher implementation as a self-synchronizing stream cipher?

- A. Cipher Block Chaining Mode,
- B. Cipher Feedback Mode
- C. Electronic Codebook Mode
- D. Output Feedback Mode

ugcnetcse-oct2022-paper1

Answer key

33.0.80 UGC NET CSE | October 2022 | Part 1 | Question: 24



Given below are two statements :

Statement I: Consider 20 bit 'Branch' microinstruction code format given below:

3	3	3	2	2	7
F1	F2	F3	CD	BR	AD

F1, F2, F3: Micro-operation fields

CD : Condition for branching

BR : Branch field

AD: Address field

Statement II: Instruction represented in above format can perform branch in

In the light of the above statements, choose the most appropriate answer from the options given below :

- A. Both Statement I and Statement II are correct
- B. Both Statement I and Statement II are incorrect
- C. Statement I is correct but Statement II is incorrect
- D. Statement I is incorrect but Statement II is correct

ugcnetcse-oct2022-paper1

33.0.81 UGC NET CSE | October 2022 | Part 1 | Question: 23



Consider the following NPDA = $(\{q_0, q_1, q_f\} \{a, b\}, \{1, z\}, \delta, q_0, z, \{q_f\})$

$$\delta(q_0, \lambda, z) = \{(q_f, z)\}$$

$$\delta(q_0, a, z) = \{(q_1, 11z)\}$$

$$\delta(q_1, a, 1) = \{(q_1, 111)\}$$

$$\delta(q_1, b, 1) = \{(q_1, \lambda)\}$$

$$\delta(q_1, \lambda, z) = \{(q_f, z)\}$$

Which of the following Language L is accepted by NPDA?

- A. $L = \{a^{2n}b^n : n \geq 0\}$
- B. $L = \{a^n b^{2n} : n \geq 0\}$
- C. $L = \{a^{2n}b^n : n > 0\}$
- D. $L = \{a^n b^{2n} : n > 0\}$

ugcnetcse-oct2022-paper1

Answer key

33.0.82 UGC NET CSE | October 2022 | Part 1 | Question: 22



Consider the language $L = \{a^n b^m : n \geq 4, m \leq 3\}$

Which of the following regular expression represents language L ?

- A. $aaaa^*(\lambda + b + bb + bbb)$
- B. $aaaaa^*(b + bb + bbb)$
- C. $aaaaa^*(\lambda + b + bb + bbb)$
- D. $aaaa^*(b + bb + bbb)$

ugcnetcse-oct2022-paper1

Answer key

33.0.83 UGC NET CSE | October 2022 | Part 1 | Question: 21



Consider a memory system having address spaced at a distance of m , T = Bank cycle time and n number of banks, then the average data access time per word access in synchronous organization is

- A. $t = \begin{cases} m, \frac{T}{n} & \text{for } m \ll n \\ T & \text{for } m \gg n \end{cases}$

- B. $t = \begin{cases} T/n & \text{for } m \ll n \\ T & \text{for } m \gg n \end{cases}$
- C. $t = \begin{cases} m \cdot T & \text{for } m \ll n \\ T & \text{for } m \gg n \end{cases}$
- D. $t = \begin{cases} m \cdot T & \text{for } m \ll n \\ m/T & \text{for } m \gg n \end{cases}$

ugcnetcse-oct2022-paper1

Answer key 

33.0.84 UGC NET CSE | October 2022 | Part 1 | Question: 20



Which layer divides each message into packets at the source and re-assembles them at the destination?

- A. Network layer
C. Data link layer
- B. Transport layer
D. Physical layer

ugcnetcse-oct2022-paper1

Answer key 

33.0.85 UGC NET CSE | October 2022 | Part 1 | Question: 19



Which of the following is an indirect measure of product?

- A. Quality
B. Complexity
C. Reliability
D. All of these

ugcnetcse-oct2022-paper1

Answer key 

33.0.86 UGC NET CSE | October 2022 | Part 1 | Question: 18



Consider a B-tree of height h , minimum degree $t \geq 2$ that contains any n -key, where $n \geq 1$. Which of the following is correct?

- A. $h \geq \log_t \frac{n+1}{2}$
C. $h \geq \log_t \frac{n-1}{2}$
- B. $h \leq \log_t \frac{n+1}{2}$
D. $h \leq \log_t \frac{n-1}{2}$

ugcnetcse-oct2022-paper1

33.0.87 UGC NET CSE | October 2022 | Part 1 | Question: 17



A classless address is given as 167.199.170.82/27. The number of addresses in the network is

- A. 64 addresses
B. 32 addresses
C. 28 addresses
D. 30 addresses

ugcnetcse-oct2022-paper1

Answer key 

33.0.88 UGC NET CSE | October 2022 | Part 1 | Question: 16



Fault base testing technique is

- A. Unit testing
C. Stress testing
- B. Beta testing,
D. Mutation testing

ugcnetcse-oct2022-paper1

Answer key 

33.0.89 UGC NET CSE | October 2022 | Part 1 | Question: 15



Assume that $f(n)$ and $g(n)$ are asymptotically positive. Which of the following is correct?

- A. $f(n) = O(g(n))$ and $g(n) = O(h(n)) \Rightarrow f(n) = \omega(h(n))$
- B. $f(n) = \Omega(g(n))$ and $g(n) = \Omega(h(n)) \Rightarrow f(n) = O(h(n))$

- C. $f(n) = o(g(n))$ and $g(n) = o(h(n)) \Rightarrow f(n) = o(h(n))$
 D. $f(n) = \omega(g(n))$ and $g(n) = \omega(h(n)) \Rightarrow f(n) = \Omega(h(n))$

ugcnetcse-oct2022-paper1

Answer key 

33.0.90 UGC NET CSE | October 2022 | Part 1 | Question: 14



Which Metrics are derived by normalizing quality and/or productivity measures by considering the size of the software that has been produced?

- A. Function - Oriented Metrics
 B. Function - Point Metrics
 C. Line of Code Metrics
 D. Size Oriented Metrics

ugcnetcse-oct2022-paper1

Answer key 

33.0.91 UGC NET CSE | October 2022 | Part 1 | Question: 12



Consider the hash table of size 11 that uses open addressing with linear probing. Let $h(k) = k \bmod 11$ be the hash function. A sequence of records with keys 43, 36, 92, 87, 11, 47, 11, 13, 14 is inserted into an initially empty hash table, the bins of which are indexed from 0 to 10. What is the index of the bin into which the last record is inserted?

- A. 8
 B. 7
 C. 10
 D. 4

ugcnetcse-oct2022-paper1

Answer key 

33.0.92 UGC NET CSE | October 2022 | Part 1 | Question: 11



In reference to Big data, consider the following database :

- (A) Memcached
 (B) Couch DB
 (C) Infinite graph

Choose the most appropriate answer from the options given below :

- A. (A) and (B) only
 B. (B) and (C) only,
 C. (C) and (A) only
 D. (A) (B) and (C)

ugcnetcse-oct2022-paper1

Answer key 

33.0.93 UGC NET CSE | October 2022 | Part 1 | Question: 10



Consider the following statements:

Statement I: Conservative 2PL is a deadlock-free protocol.

Statement II: Thomas's write rule enforces conflict serializability.

Statement III: Timestamp ordering protocol ensures serializability based on the order of transaction timestamps.

Which of the following is correct?

- A. Statement I, Statement II true and Statement III false
 B. Statement I, Statement III true and Statement II false
 C. Statement I, Statement II false and Statement III true
 D. Statement I, Statement II and Statement III true

ugcnetcse-oct2022-paper1

Answer key 

33.0.94 UGC NET CSE | October 2022 | Part 1 | Question: 9

The number of nodes of height h in any n -element heap is atmost:

- A. $n/2^{h+1}$ B. $\frac{n}{2^{h-1}}$ C. $\frac{n}{2^h}$ D. $\frac{n-1}{2^{h-1}}$

ugcnetcse-oct2022-paper1

33.0.95 UGC NET CSE | October 2022 | Part 1 | Question: 8

A 4-stage pipeline has the stage delay as 150, 120, 160 and 140 ns respectively. Registers that are used between the stages have delay of 5 ns. Assuming constant locking rate, the total time required to process 1000 data items on this pipeline is

- A. 160.5 ms B. 165.5 ms
C. 120.5 ms D. 590.5 ms

ugcnetcse-oct2022-paper1

Answer key

33.0.96 UGC NET CSE | October 2022 | Part 1 | Question: 7

Match List I with List II :

List I	List II
(A) Ontological Engineering	(I) Organizing subclass relations
(B) Taxonomy Hierarchy	(II) Organizing knowledge into category and sub category
(C) Inheritance	(III) Attaches a number with each possibility
(D) Probability mode	(IV) Representing concepts. events, time. physical concepts of different domains

Choose the correct answer from the options given below :

- A. (A)-(II), (B)-(I), (C)-(IV), (D)-(III) B. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
C. (A)-(IV), (B)-(III), (C)-(I), (D)-(II) D. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

ugcnetcse-oct2022-paper1

Answer key

33.0.97 UGC NET CSE | October 2022 | Part 1 | Question: 6

The solution of the recurrence relation $T(n) = 3T(n/4) + n \lg n$ is

- A. $\theta(n^2 \lg n)$ B. $\theta(n \lg n)$
C. $\theta(n \lg n)^2$ D. $\theta(n \lg \lg n)$

ugcnetcse-oct2022-paper1

33.0.98 UGC NET CSE | October 2022 | Part 1 | Question: 5

Size and complexity are a part of

- A. People Metrics B. Project Metrics
C. Process Metrics D. Product Metrics

ugcnetcse-oct2022-paper1

Answer key

33.0.99 UGC NET CSE | October 2022 | Part 1 | Question: 4

Consider the following statements about Context Free Language (CFL) :

Statement I: CFL is closed under homomorphism.

Statement II: CFL is closed under complement.

Which of the following is correct?

- A. Statement I is true and Statement II is false
- B. Statement II is true and Statement I is false
- C. Both Statement I and Statement II are true
- D. Neither Statement I nor Statement II is true

ugcnetcse-oct2022-paper1

Answer key

33.0.100 UGC NET CSE | October 2022 | Part 1 | Question: 3

Using 'RSA' algorithm, if $p = 13$, $q = 5$ and $e = 7$. the value of d and cipher value of '6' with (e, n) key are

- A. 7,4
- B. 7,1
- C. 7,46
- D. 55,1

ugcnetcse-oct2022-paper1

33.0.101 UGC NET CSE | October 2022 | Part 1 | Question: 2

Consider two lists A and B of three strings on $\{0, 1\}$

X :

List A	List B
1	111
10111	10
10	0

Y:

List A	List B
10	101
011	11
101	011

Which of the following is true?

- A. Only PCP in X has solution.,
- B. Only PCP in Y has solution.,
- C. PCP in both X and Y has solution.
- D. PCP neither in X nor in Y has solution.,

ugcnetcse-oct2022-paper1

Answer key

33.1 3 Puzzle (1)**33.1.1 3 Puzzle: UGC NET CSE | July 2016 | Part 3 | Question: 74**

Consider a 3-puzzle where, like in the usual 8-puzzle game, a tile can only move to an adjacent empty space. Given the initial state

1	2
	3

which of the following state cannot be reached?

A	3	1
		2
B		3
	2	1
C	1	3
		2
D		2
	1	3

ugcnetcse-july2016-paper3 artificial-intelligence 3-puzzle

Answer key 

33.2

Algorithm Design (1)

33.2.1 Algorithm Design: UGC NET CSE | October 2022 | Part 1 | Question: 74



Which of the following algorithm design approach is used in Quick sort algorithm?

- A. Dynamic programming
- B. Back Tracking
- C. Divide and conquer
- D. Greedy approach

ugcnetcse-oct2022-paper1 algorithm-design quick-sort

Answer key 

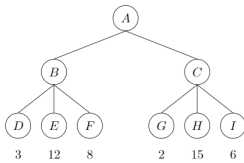
33.3

Alpha Beta Pruning (1)

33.3.1 Alpha Beta Pruning: UGC NET CSE | July 2016 | Part 3 | Question: 73



Consider the following game tree in which root is a maximizing node and children are visited left to right. What nodes will be pruned by the alpha-beta pruning?



- A. *I*
- B. *HI*
- C. *CHI*
- D. *GHI*

ugcnetcse-july2016-paper3 artificial-intelligence alpha-beta-pruning

Answer key 

33.4

Alpha Cut (1)

33.4.1 Alpha Cut: UGC NET CSE | June 2019 | Part 2 | Question: 95



Let A_{α_0} denotes the α -cut of a fuzzy set A at α_0 . If $\alpha_1 < \alpha_2$, then

- A. $A_{\alpha_1} \supseteq A_{\alpha_2}$
- B. $A_{\alpha_1} \supset A_{\alpha_2}$
- C. $A_{\alpha_1} \subseteq A_{\alpha_2}$
- D. $A_{\alpha_1} \subset A_{\alpha_2}$

ugcnetcse-june2019-paper2 fuzzy-set alpha-cut

Answer key 

33.5

Application Software (1)

33.5.1 Application Software: UGC NET CSE | June 2010 | Part 2 | Question: 45



Applications-software

- A. Is used to control the operating system
- B. Includes programs designed to help programmers

C. Performs a specific task for computer users

D. All of the above

ugcnetcse-june2010-paper2 application-software

Answer key 

33.6

Bigdata Hadoop (1)

33.6.1 Bigdata Hadoop: UGC NET CSE | June 2019 | Part 2 | Question: 38



Hadoop (a big data tool) works with number of related tools. Choose from the following, the common tools included into Hadoop:

- A. MySQL, Google API and Map reduce
- C. Map reduce, H Base and Hive

- B. Map reduce, Scala and Hummer
- D. Map reduce, Hummer and Heron

ugcnetcse-june2019-paper2 bigdata-hadoop

Answer key 

33.7

Convex Optimization (1)

33.7.1 Convex Optimization: UGC NET CSE | July 2018 | Part 2 | Question: 82



Which of the following statements is false about convex minimization problem?

- A. If a local minimum exists, then it is a global minimum
- B. The set of all global minima is convex set
- C. The set of all global minima is concave set
- D. For each strictly convex function, if the function has a minimum, then the minimum is unique

ugcnetcse-july2018-paper2 convex-optimization

Answer key 

33.8

Data Mining (2)

33.8.1 Data Mining: UGC NET CSE | July 2016 | Part 2 | Question: 50



Given the following two statements:

- I. Strategic value of data mining is timestamping
- II. Information collection is an expensive process in building an expert system

Which of the following options is correct?

- A. Both I and II are false
- B. Both I and II are true
- C. I is true, II is false
- D. I is false, II is true

ugcnetcse-july2016-paper2 data-mining

Answer key 

33.8.2 Data Mining: UGC NET CSE | June 2010 | Part 2 | Question: 50



Data Mining uses _____, _____ and _____ to build effective predictive model.

- (i) Data set
- (ii) Information set
- (iii) Input set
- (iv) Process set
- (v) Output set

(vi) Test set

(i), (ii) and (iv)

(ii), (iv) and (v)

(i), (v) and (vi)

(ii), (iii) and (v)

ugcnetcse-june2010-paper2 data-mining

33.9

Data Warehousing (4)

33.9.1 Data Warehousing: UGC NET CSE | July 2016 | Part 2 | Question: 47



Consider the following two statements:

- I. Business intelligence and Data warehousing is used for forecasting and Data mining
- II. Business intelligence and Data warehousing is used for analysis of large volumes of sales data

Which one of the following options is correct?

- A. I is true , II is false
- B. Both I and II are true
- C. I is false II is false
- D. Both I and II are false

ugcnetcse-july2016-paper2 data-warehousing

Answer key

33.9.2 Data Warehousing: UGC NET CSE | July 2016 | Part 2 | Question: 49



Consider the following two statements:

- I. Data scrubbing is a process to upgrade the quality of data, before it is moved into Data warehouse
- II. Data scrubbing is a process of rejecting data from data warehouse to create indexes

Which one of the following options is correct?

- A. I is true II is false
- B. I is false II is true
- C. Both I and II are false
- D. Both I and II are true

ugcnetcse-july2016-paper2 data-warehousing

Answer key

33.9.3 Data Warehousing: UGC NET CSE | June 2012 | Part 3 | Question: 20



Data Warehouse provides

- A. Transaction Responsiveness
- B. Storage, Functionality Responsiveness to queries
- C. Demand and Supply Responsiveness
- D. None of the above

ugcnetcse-june2012-paper3 data-warehousing

Answer key

33.9.4 Data Warehousing: UGC NET CSE | June 2012 | Part 3 | Question: 7



Match the following :

- (i) OLAP
- (ii) OLTP
- (iii) Decision tree
- (iv) Neural Network
- (a) Regression
- (b) Data Warehouse
- (c) RDBMS
- (d) Classification

- A. (i)-(b), (ii)-(c), (iii)-(a), (iv)-(d)
- B. (i)-(b), (ii)-(c), (iii)-(d), (iv)-(a)
- C. (i)-(c), (ii)-(b), (iii)-(a), (iv)-(d)
- D. (i)-(b), (ii)-(c), (iii)-(d), (iv)-(a)

Answer key 

33.10

Debugger (1)

33.10.1 Debugger: UGC NET CSE | December 2014 | Part 2 | Question: 34



Debugger is a program that

- A. Allows to examine and modify the contents of registers
- B. Does not allow execution of a segment of program
- C. Allows to set breakpoints, execute a segment of program and display contents of register
- D. All of the above

ugcnetcse-dec2014-paper2 debugger

Answer key 

33.11

Defuzzification (1)

33.11.1 Defuzzification: UGC NET CSE | June 2019 | Part 2 | Question: 96



Consider the following methods:

M_1 : Mean of maximum

M_2 : Centre of area

M_3 : Height method

Which of the following is/are defuzzification method(s)?

- A. Only M_2
- B. Only M_1 and M_2
- C. Only M_2 and M_3
- D. M_1 , M_2 and M_3

ugcnetcse-june2019-paper2 fuzzy-set defuzzification

Answer key 

33.12

Encryption Decryption (1)

33.12.1 Encryption Decryption: UGC NET CSE | July 2018 | Part 2 | Question: 48



Decrypt the message "WTAAD" using the Caesar Cipher with key =15

- A. LIPPS
- B. HELLO
- C. PLLEH
- D. DAATW

ugcnetcse-july2018-paper2 cryptography encryption-decryption

Answer key 

33.13

Fuzzy Relation (1)

33.13.1 Fuzzy Relation: UGC NET CSE | July 2016 | Part 3 | Question: 64



Let R and S be two fuzzy relations defined as

$$R = \begin{matrix} & y_1 & y_2 \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.7 & 0.5 \\ 0.8 & 0.4 \end{bmatrix} \end{matrix}$$

and

$$S = \begin{matrix} & z_1 & z_2 & z_3 \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 0.9 & 0.6 & 0.2 \\ 0.1 & 0.7 & 0.5 \end{bmatrix} \end{matrix}$$

Then, the resulting relation T, which related elements of universe x to elements of universe z using max-min composition is given by

- A. $T = x_1 \begin{bmatrix} z_1 & z_2 & z_3 \\ .5 & .7 & .5 \end{bmatrix}$
 $x_2 \begin{bmatrix} .8 & .8 & .8 \end{bmatrix}$
- B. $T = x_1 \begin{bmatrix} z_1 & z_2 & z_3 \\ .5 & .7 & .5 \end{bmatrix}$
 $x_2 \begin{bmatrix} .9 & .6 & .5 \end{bmatrix}$
- C. $T = x_1 \begin{bmatrix} z_1 & z_2 & z_3 \\ 0.7 & 0.6 & 0.5 \end{bmatrix}$
 $x_2 \begin{bmatrix} 0.8 & 0.6 & 0.4 \end{bmatrix}$
- D. $T = x_1 \begin{bmatrix} z_1 & z_2 & z_3 \\ 0.7 & 0.6 & 0.5 \end{bmatrix}$
 $x_2 \begin{bmatrix} 0.8 & 0.8 & 0.8 \end{bmatrix}$

ugcnetcse-july2016-paper3 fuzzy-set fuzzy-relation

Answer key 

33.14

Fuzzy Set (5)

33.14.1 Fuzzy Set: UGC NET CSE | December 2012 | Part 3 | Question: 13



Consider a fuzzy set A defined on the interval $X=[0, 10]$ of integers by the membership function $\mu_A(x) = \frac{x}{x+2}$. Then the α cut corresponding to $\alpha=0.5$ will be

- A. $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ B. $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
C. $\{2, 3, 4, 5, 6, 7, 8, 9, 10\}$ D. $\{\}$

ugcnetcse-dec2012-paper3 fuzzy-set

Answer key 

33.14.2 Fuzzy Set: UGC NET CSE | December 2014 | Part 3 | Question: 71



A _____ point of a fuzzy set A is a point $x \in X$ at which $\mu_A(x) = 0.5$

- A. Core B. Support C. Crossover D. α -cut

ugcnetcse-dec2014-paper3 fuzzy-set

Answer key 

33.14.3 Fuzzy Set: UGC NET CSE | December 2015 | Part 3 | Question: 65



Consider a standard additive model consisting of rules of the form of

If x is A_i AND y is B_i THEN z is C_i . Given crisp inputs $x = x_0$, $y = y_0$ the output of the model is

- A. $z = \sum_i \mu_{A_i}(x_0) \mu_{B_i}(y_0) \mu_{C_i}(z)$
B. $z = \sum_i \mu_{A_i}(x_0) \mu_{B_i}(y_0)$
C. $z = \text{centroid}(\sum_i \mu_{A_i}(x_0) \mu_{B_i}(y_0) \mu_{C_i}(z))$
D. $z = \text{centroid}(\sum_i \mu_{A_i}(x_0) \mu_{B_i}(y_0))$

ugcnetcse-dec2015-paper3 fuzzy-set

Answer key 

33.14.4 Fuzzy Set: UGC NET CSE | December 2015 | Part 3 | Question: 66



A bell shaped membership function is specified by three parameters (a, b, c) as follows:

- A. $\frac{1}{1 + \left(\frac{x-c}{a}\right)^b}$ B. $\frac{1}{1 + \left(\frac{x-c}{a}\right)^{2b}}$

C.

$$1 + \left(\frac{x-c}{a} \right)^b$$

D. $1 + \left(\frac{x-c}{a} \right)^{2b}$

ugcnetcse-dec2015-paper3 fuzzy-set

Answer key **33.14.5 Fuzzy Set: UGC NET CSE | July 2016 | Part 3 | Question: 65**

Compute the value of adding the following two fuzzy integers:

$$A = \{(0.3, 1), (0.6, 2), (1, 3), (0.7, 4), (0.2, 5)\}$$

$$B = \{(0.5, 11), (1, 12), (0.5, 13)\}$$

Where fuzzy addition is defined as

$$\mu_{A+B}(z) = \max_{x+y=z} (\min(\mu_A(x), \mu_B(y)))$$

Then, $f(A+B)$ is equal to

- A. $\{(0.5, 12), (0.6, 13), (1, 14), (0.7, 15), (0.7, 16), (1, 17), (1, 18)\}$
- B. $\{(0.5, 12), (0.6, 13), (1, 14), (1, 15), (1, 16), (1, 17), (1, 18)\}$
- C. $\{(0.3, 12), (0.5, 13), (0.5, 14), (1, 15), (0.7, 16), (0.5, 17), (0.2, 18)\}$
- D. $\{(0.3, 12), (0.5, 13), (0.6, 14), (1, 15), (0.7, 16), (0.5, 17), (0.2, 18)\}$

ugcnetcse-july2016-paper3 fuzzy-set

Answer key **33.15****Geographical Information System (1)****33.15.1 Geographical Information System: UGC NET CSE | December 2011 | Part 2 | Question: 46**

To compare, overlay or cross analyze to maps in *GIS*.

- A. Both maps must be in digital form
- B. Both maps must be at the same equivalent scale.
- C. Both maps must be on the same coordinate system
- D. All of the above

ugcnetcse-dec2011-paper2 geographical-information-system

Answer key **33.16****Hill Climbing Search (1)****33.16.1 Hill Climbing Search: UGC NET CSE | July 2016 | Part 3 | Question: 72**

How does randomized Hill climbing choose the next move each time?

- A. It generates a random move from moveset and accepts this move
- B. It generates random move from whole state space, and accepts this move
- C. It generates a random move from moveset and accepts this move only if this move improves the evaluation function
- D. It generates a random move from whole state space, and accepts this move only if this move improves the evaluation function

ugcnetcse-july2016-paper3 artificial-intelligence hill-climbing-search

Answer key **33.17****Inventory (1)****33.17.1 Inventory: UGC NET CSE | December 2014 | Part 2 | Question: 49**

_____ model is designed to bring prices down by increasing the number of customers who buy a particular product at once.

- A. Economic Order Quantity
- B. Inventory

Answer key 

33.18

K Mean Clustering (1)

33.18.1 K Mean Clustering: UGC NET CSE | June 2019 | Part 2 | Question: 40



K-mean clustering algorithm has clustered the given 8 observations into 3 clusters after 1st iteration as follows:

$$C1 : \{(3, 3), (5, 5), (7, 7)\}$$

$$C2 : \{(0, 6), (6, 0), (3, 0)\}$$

$$C3 : \{(8, 8), (4, 4)\}$$

What will be the Manhattan distance for observation $(4, 4)$ from cluster centroid $C1$ in the second iteration?

- A. 2 B. $\sqrt{2}$ C. 0 D. 18

ugcnetcse-june2019-paper2 k-mean-clustering data-mining

Answer key 

33.19

Llp (1)

33.19.1 Llp: UGC NET CSE | July 2018 | Part 2 | Question: 83



The following LLP

$$\text{Maximize } z = 100x_1 + 2x_2 + 5x_3$$

Subject to

$$14x_1 + x_2 - 6x_3 + 3x_4 = 7$$

$$32x_1 + x_2 - 12x_3 \leq 10$$

$$3x_1 - x_2 - x_3 \leq 0$$

$$x_1, x_2, x_3, x_4 \geq 0 \text{ has}$$

- A. Solution : $x_1 = 100, x_2 = 0, x_3 = 0$ B. Unbounded solution
C. No solution D. Solution : $x_1 = 50, x_2 = 70, x_3 = 60$

ugcnetcse-july2018-paper2 llp linear-programming

Answer key 

33.20

Machine Learning (1)

33.20.1 Machine Learning: UGC NET CSE | June 2014 | Part 2 | Question: 29



Match the following :

List – I**List – II**

- | | |
|-----------------------|---------------------------------|
| a. Classification | i. Principal component analysis |
| b. Clustering | ii. Branch and Bound |
| c. Feature Extraction | iii. K-nearest neighbour |
| d. Feature Selection | iv. K-means |

Codes :

- A. a-iii; b-iv; c-ii; d-i B. a-iv; b-iii; c-i; d-ii
C. a-iii; b-iv; c-i; d-ii D. a-iv; b-iii; c-ii; d-i

ugcnetcse-june2014-paper2 machine-learning

Answer key 

33.21

Net Exam (1)

33.21.1 Net Exam: Information regarding UGC NET



Can someone please tell what's the use of giving UGC NET in computer science after doing MTech and have given GATE in CSE?

What benefits will NET give when applying for Phd or as an assistant professor?

What is the requirement of NET? I am currently in 1st year MTech, AFAIK we are eligible only if we are in final year MTECH(ie 2nd year) is this right?

net-exam general admissions query

33.22

Operational Research (2)

33.22.1 Operational Research: UGC NET CSE | December 2014 | Part 3 | Question: 68



The occurrence of degeneracy while solving a transportation problem means that

- A. Total supply equals total demand
- B. Total supply does not equal total demand
- C. The solution so obtained is not feasible
- D. None of these

ugcnetcse-dec2014-paper3 operational-research transportation-problem

Answer key

33.22.2 Operational Research: UGC NET CSE | September 2013 | Part 3 | Question: 14



An optimal assignment requires that the minimum number of horizontal and vertical lines that can be drawn to cover all zeros be equal to the number of

- A. rows or columns
- B. rows + columns
- C. rows + columns -1
- D. rows + columns +1

ugcnetcse-sep2013-paper3 operational-research assignment-problem

Answer key

33.23

Optimization (1)

33.23.1 Optimization: UGC NET CSE | July 2016 | Part 3 | Question: 61



The region of feasible solution of a linear programming problem has a ____ property in geometry, provided the feasible solution of the problem exists

- A. concavity
- B. convexity
- C. quadratic
- D. polyhedron

ugcnetcse-july2016-paper3 optimization linear-programming

Answer key

33.24

Process Scheduling (1)

33.24.1 Process Scheduling: UGC NET CSE | October 2022 | Part 1 | Question: 60



What is called Journalling in Linux operating system?

- A. Process scheduling
- B. File saving as transaction,
- C. A type of thread
- D. An editor,

ugcnetcse-oct2022-paper1 operating-system process-scheduling

Answer key

33.25

Searching (1)

33.25.1 Searching: UGC NET CSE | December 2014 | Part 3 | Question: 56



An A* algorithm is a heuristic search technique which

- A. Is like a depth-first search where most promising child is selected for expansion
- B. Generates all successor nodes and computes an estimate of distance (cost) from start node to a goal node through each of the successors. It then chooses the successor with shortest cost.
- C. Saves all path lengths (costs) from start node to all generated nodes and chooses shortest path for further expansion.
- D. None of the above

ugcnetcse-dec2014-paper3 artificial-intelligence searching

Answer key 

33.26 Shell (1)

33.26.1 Shell: UGC NET CSE | June 2012 | Part 3 | Question: 71



Which one of the following is not a shell in UNIX system?

- A. Bourne Shell
- B. C Shell
- C. Not Shell
- D. Korn Shell

ugcnetcse-june2012-paper3 unix shell

Answer key 

33.27 Shell Script (2)

33.27.1 Shell Script: UGC NET CSE | June 2014 | Part 2 | Question: 35



Which of the following commands will output “onetwothree” ?

- A. for val; do echo-n \$val; done < one two three
- B. for one two three; do echo-n-; done
- C. for n in one two three; do echo-n \$n; done
- D. for n in one two three {echo -n \$ n}

ugcnetcse-june2014-paper2 non-gatecse unix shell-script

Answer key 

33.27.2 Shell Script: UGC NET CSE | June 2014 | Part 3 | Question: 39



Match the following with reference to Unix shell scripts :

List – I

List – II

- | | |
|--------|------------------------------------|
| a. \$? | i. File name of the current script |
| b. \$# | ii. List of arguments |
| c. \$0 | iii. The number of arguments |
| d. \$* | iv. Exit status of last command |

Codes :

- | | |
|---------------------------|---------------------------|
| A. a-iii, b-ii, c-i, d-iv | B. a-ii, b-iii, c-i, d-iv |
| C. a-iv, b-iii, c-i, d-ii | D. a-i, b-iii, c-i, d-iv |

ugcnetjune2014iii unix shell-script

Answer key 

33.28 Simplex (1)

33.28.1 Simplex: UGC NET CSE | October 2022 | Part 1 | Question:: 13



Consider the primal problem :

Maximize $z = 5x_1 + 12x_2 + 4x_3$

Subject to

$$\begin{aligned}x_1 + 2x_2 + x_3 &= 10 \\2x_1 - x_2 + 3x_3 &= 8 \\x_1, x_2, x_3 &\geq 0\end{aligned}$$

its dual problem is
Minimize $w = 10y_1 + 8y_2$
Subject to

$$\begin{aligned}y_1 + 2y_2 &\geq 5 \\2y_1 - y_2 &\geq 12 \\y_1 + 3y_2 &\geq 4\end{aligned}$$

Which of the following is correct?

- A. $y_1 \geq 0, y_2$ unrestricted
C. y_1 is unrestricted, $y_2 \geq 0$
B. $y_1 \geq 0, y_2 \geq 0$
D. y_1 is unrestricted, y_2 restricted

ugcnetcse-oct2022-paper1 simplex

Answer key 

33.29

Simplex Method (2)

33.29.1 Simplex Method: UGC NET CSE | December 2014 | Part 3 | Question: 67



If an artificial variable is present in the 'basic variable' column of optimal simplex table, then the solution is

- A. Optimum
C. Unbounded
B. Infeasible
D. Degenerate

ugcnetcse-dec2014-paper3 linear-programming simplex-method

Answer key 

33.29.2 Simplex Method: UGC NET CSE | July 2016 | Part 3 | Question: 62



Consider the following statements:

- Revised simplex method requires lesser computations than the simplex methods
- Revised simplex method automatically generates the inverse of the current basis matrix
- Less number of entries are needed in each table of the revised simplex method than usual simplex method

Which of these statements is true?

- A. i and ii only
C. ii and iii only
B. i and iii only
D. i, ii, and iii

ugcnetcse-july2016-paper3 optimization simplex-method

Answer key 

33.30

System Software& Compilers (1)

33.30.1 System Software& Compilers: UGC NET CSE | July 2016 | Part 2 | Question: 32



The content of the accumulator after the execution of the following 8085 assembly language program, is:

```
MVI    A,
42H
MVI B, 05h
UGC: ADD B
DCR B
JNZ UGC
ADI 25H
```

HLT

A. 82 H

B. 78 H

C. 76 H

D. 47 H

ugcnetcse-july2016-paper2 system-software&-compilers

Answer key 

33.31

Transportation Problem (1)

33.31.1 Transportation Problem: UGC NET CSE | July 2016 | Part 3 | Question: 63



The following transportation problem :

	A	B	C	Supply
I	50	30	220	1
II	90	45	170	3
III	250	200	50	4
Demand	4	2	2	0

Has a solution

	A	B	C
I	1		
II	3	0	
III		2	2

The above solution of a given transportation problem is

A. Infeasible solution

B. optimum solution

C. non-optimum solution

D. unbounded solution

ugcnetcse-july2016-paper3 optimization transportation-problem

Answer key 

33.32

Unified Modeling Language (1)

33.32.1 Unified Modeling Language: UGC NET CSE | June 2012 | Part 3 | Question: 44



Which diagram provides a formal graphic notation for modelling objects, classes and their relationships to one another?

A. Object diagram

B. Class diagram

C. Instance diagram

D. Analysis diagram

ugcnetcse-june2012-paper3 unified-modeling-language

Answer key 

33.33

Unix (4)

33.33.1 Unix: UGC NET CSE | July 2016 | Part 2 | Question: 39



In UNIX, ____ creates three subdirectories; 'PIS' and two subdirectories 'progs' and 'data' from just created subdirectory 'PIS'

A. mkdir PIS/progs PIS/data PIS

B. mkdir PIS progs data

C. mkdir PIS PIS/progs PIS/data

D. mkdir PIS/progs data

ugcnetcse-july2016-paper2 unix

Answer key 

33.33.2 Unix: UGC NET CSE | July 2016 | Part 3 | Question: 52



Which of the following option with reference to UNIX operating system is not correct?

A. INT signal is sent by the terminal driver when one types <Control-C> and it is a request to terminate the current

operation

- B. TERM is a request to terminate execution completely. The receiving process will clean up its state and exit
- C. QUIT is similar to TERM, except that it defaults to producing a core dump if not caught.
- D. KILL is a blockable signal

ugcnetcse-july2016-paper3 unix

Answer key 

33.33.3 Unix: UGC NET CSE | July 2016 | Part 3 | Question: 67



What is the function of following UNIX command?

WC -l<a >b &

- A. It runs the word count program to count the number of lines in its input, a, writing the result to b, as a foreground process
- B. It runs the word count program to count the number of lines in its input, a, writing the result to b, but does it in the background
- C. It counts the error during the execution of the process, a, and puts the result in process b
- D. It copies the 'l' numbers of lines of program file, a, and stores in file b

ugcnetcse-july2016-paper3 unix

Answer key 

33.33.4 Unix: UGC NET CSE | June 2010 | Part 2 | Question: 49



At any given time Parallel Virtual Machine (PVM) has _____ send buffer and _____ receive buffer.

- A. One-one
- B. One-two
- C. Two-two
- D. Two-one

ugcnetcse-june2010-paper2 unix

Answer key 

33.34

Vertex Cover (1)

33.34.1 Vertex Cover: UGC NET CSE | October 2022 | Part 1 | Question: 56



Consider the following statements of approximation algorithm :

Statement I: Vertex-cover is a polynomial time 2-approximation algorithm.

Statement II: TSP-tour is a polynomial time 3-approximation algorithm for travelling salesman problem with the triangle inequality.

Which of the following is correct?

- A. Statement I true and Statement II false
- B. Statement I and Statement II true
- C. Statement I false and Statement II true
- D. Statement I and Statement II false

ugcnetcse-oct2022-paper1 algorithms vertex-cover

33.35

Windows (1)

33.35.1 Windows: UGC NET CSE | June 2012 | Part 3 | Question: 69



What is the meaning of 'Hibernate' in WindowsXP/Windows 7?

- A. Restart the computers in safe mode
- B. Restart the computers in normal mode
- C. Shutdown the computer terminating all the running applications

D. Shutdown the computer without closing the running applications

ugcnetcse-june2012-paper3 windows

Answer key 

Answer Keys

33.0.1	Q-Q	33.0.2	Q-Q	33.0.3	Q-Q	33.0.4	Q-Q	33.0.5	Q-Q
33.0.6	Q-Q	33.0.7	Q-Q	33.0.8	Q-Q	33.0.9	Q-Q	33.0.10	Q-Q
33.0.11	Q-Q	33.0.12	Q-Q	33.0.13	Q-Q	33.0.14	Q-Q	33.0.15	Q-Q
33.0.16	Q-Q	33.0.17	Q-Q	33.0.18	Q-Q	33.0.19	Q-Q	33.0.20	Q-Q
33.0.21	Q-Q	33.0.22	Q-Q	33.0.23	Q-Q	33.0.24	Q-Q	33.0.25	Q-Q
33.0.26	Q-Q	33.0.27	Q-Q	33.0.28	Q-Q	33.0.29	Q-Q	33.0.30	Q-Q
33.0.31	Q-Q	33.0.32	Q-Q	33.0.33	Q-Q	33.0.34	Q-Q	33.0.35	Q-Q
33.0.36	Q-Q	33.0.37	Q-Q	33.0.38	Q-Q	33.0.39	Q-Q	33.0.40	Q-Q
33.0.41	Q-Q	33.0.42	Q-Q	33.0.43	Q-Q	33.0.44	Q-Q	33.0.45	Q-Q
33.0.46	Q-Q	33.0.47	Q-Q	33.0.48	Q-Q	33.0.49	Q-Q	33.0.50	Q-Q
33.0.51	Q-Q	33.0.52	Q-Q	33.0.53	Q-Q	33.0.54	Q-Q	33.0.55	Q-Q
33.0.56	Q-Q	33.0.57	Q-Q	33.0.58	Q-Q	33.0.59	Q-Q	33.0.60	Q-Q
33.0.61	Q-Q	33.0.62	Q-Q	33.0.63	Q-Q	33.0.64	Q-Q	33.0.65	Q-Q
33.0.66	Q-Q	33.0.67	Q-Q	33.0.68	Q-Q	33.0.69	Q-Q	33.0.70	Q-Q
33.0.71	Q-Q	33.0.72	Q-Q	33.0.73	Q-Q	33.0.74	Q-Q	33.0.75	Q-Q
33.0.76	Q-Q	33.0.77	Q-Q	33.0.78	Q-Q	33.0.79	Q-Q	33.0.80	Q-Q
33.0.81	Q-Q	33.0.82	Q-Q	33.0.83	Q-Q	33.0.84	Q-Q	33.0.85	Q-Q
33.0.86	Q-Q	33.0.87	Q-Q	33.0.88	Q-Q	33.0.89	Q-Q	33.0.90	Q-Q
33.0.91	Q-Q	33.0.92	Q-Q	33.0.93	Q-Q	33.0.94	Q-Q	33.0.95	Q-Q
33.0.96	Q-Q	33.0.97	Q-Q	33.0.98	Q-Q	33.0.99	Q-Q	33.0.100	Q-Q
33.0.101	Q-Q	33.1.1	C	33.2.1	Q-Q	33.3.1	B	33.4.1	A
33.5.1	Q-Q	33.6.1	C	33.7.1	Q-Q	33.8.1	B	33.8.2	Q-Q
33.9.1	B	33.9.2	A	33.9.3	B	33.9.4	B	33.10.1	C
33.11.1	D	33.12.1	Q-Q	33.13.1	C	33.14.1	C	33.14.2	C
33.14.3	D	33.14.4	A	33.14.5	D	33.15.1	Q-Q	33.16.1	C
33.17.1	D	33.18.1	A	33.19.1	Q-Q	33.20.1	C	33.21.1	Q-Q
33.22.1	C	33.22.2	A	33.23.1	B	33.24.1	Q-Q	33.25.1	B
33.26.1	C	33.27.1	C	33.27.2	C	33.28.1	Q-Q	33.29.1	B
33.29.2	D	33.30.1	C	33.31.1	B	33.32.1	A	33.33.1	C
33.33.2	D	33.33.3	B	33.33.4	Q-Q	33.34.1	Q-Q	33.35.1	D